Lesson 2: Key Press Reactor

Godot Engine

VIDEO GAME CREATION

This week: sequences, events, and the game loop. Your job is to build something that reacts when you press a key - and this is also the start of your Semester 1 project, so save it somewhere safe and give it a name you'll recognize.

Tip: this lesson's Assets folder already has a ready-to-use placeholder character image (and an optional sound effect for Level 3) - you don't need to find or make your own art today.

Setup

1. Open Godot, click New Project, name it [YourName]Game, choose a folder, Create & Edit.
2. Create a new scene, choose Node2D as the root type, rename it Main, and save it as main.tscn in the project's root folder - this is the scene that actually runs, the same role Unity's Scene, GameMaker's Room, or Scratch's Stage already play. Every other scene this course builds (Player this week, then Bullet, Enemy, and more in later lessons) gets instanced into this one.
3. Right-click main.tscn in the FileSystem dock and choose "Set As Main Scene" (or Project > Project Settings > Application > Run > Main Scene and browse to it) - this is what F5 (Run Project) launches.
4. Create a second new scene: a Sprite2D root node, rename it Player, and assign one of the placeholder images from this lesson's Assets folder (e.g. player_circle_blue.png) as its Texture.
5. Save this scene as player.tscn.
6. Back in main.tscn, drag player.tscn from the FileSystem dock into the scene tree to add Player as a child of Main.

Level 1 - Minimal (MVP)

One key press causes one visible reaction. This is a complete finish for today - reaching this is enough to move on to next week.

7. Select the Player node (open player.tscn directly, or select it inside main.tscn) and click the script icon in the Inspector, keep defaults, click Create.
8. Inside the _process(delta) function, add the code below.
9. Press F6 (Run Current Scene) with player.tscn open and hold the Right arrow key to test - F6 always runs whichever scene you currently have open, regardless of the project's Main Scene setting, so this still works for testing Player by itself.

```
func _process(delta):
    if Input.is_action_pressed("ui_right"):
        position.x += 200 * delta
```

Level 2 - Complete (Alpha/Beta)

All 4 arrow keys move your character in 4 directions.

- Add the other 3 directions using ui_left, ui_up, and ui_down (position.x -= ..., position.y -= ..., position.y += ...).

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Animate: make the sprite blink while moving, and stay solid (visible) when idle.
- Sound: play sfx_blip.wav in a beep-beep pattern while moving, silent when idle - add it via an AudioStreamPlayer2D child node.
- Recolor: swap in a different placeholder image for each direction (texture), rather than tinting one - assign the Sprite2D's texture property to a different preloaded image per direction, and back to your default/orange image when idle.
- Clamp: keep the sprite on screen using clamp() on position.
- Structure: keep all four of the above together in the same _process(delta) function, computed once per frame - not scattered across separate functions.

If Something's Wrong

- Nothing moves: confirm the script is attached to the Sprite2D that's actually in your running scene, and that you saved the scene before pressing F6.
- Red text in the Output panel points to the exact line with a typo - fix the first error listed first.
- F5 (Run Project) shows a blank window, or the old empty-project screen: confirm main.tscn was actually set as the project's Main Scene (Project Settings > Application > Run > Main Scene), and that Player was dragged into it as a child rather than left as its own separate, unconnected scene.

Want More?

docs.godotengine.org, Student Resource List > Godot